



Leonardo Maglanoc

Robot Learning & AI Engineer



Munich, Germany

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LeoMaglanoc

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- Research Interests
- Robot learning & foundation models
  - Neuro-inspired computation
  - 3D computer vision and SLAM
  - Safe robotics

Programming

Python (PyTorch, ROS2)

Linux (Git, Docker)

C++

Languages

German (mother tongue)

Norwegian (mother tongue)

English (C2)

French (B1)

Short Profile

Focusing on robot learning and foundation models. Born in Norway and raised in Germany, I'm particularly excited to contribute to Europe's deep-tech ecosystem and build ambitious robotics and AI systems here.

Publications

- J. Thumm, J. Balletshofer, L. Maglanoc, L. Muschal, and M. Althoff, A general safety framework for autonomous manipulation in human environments," *IEEE Transactions on Robotics*, 2026.

Work Experiences

|                     |                                                                                                                                                                                                                                                                      |                                                 |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Oct 2025 - May 2026 | Master Thesis Collaboration                                                                                                                                                                                                                                          | Foundation Robotics, Munich, Germany            |
|                     | Thesis on language-guided dexterous manipulation with foundation models in collaboration with a robotics startup and TUM MIRMI.                                                                                                                                      |                                                 |
| Feb 2025 - Aug 2025 | AI/ML Research Intern                                                                                                                                                                                                                                                | BMW Group Research, Munich, Germany, full-time  |
|                     | Developing neuro-inspired Oscillatory Neural Networks (ONNs) in collaboration with IBM. Designing Python/PyTorch simulators and evaluating them on data collected from a tactile hardware user study (10 participants) for a human-vehicle interaction demonstrator. |                                                 |
| Apr 2023 - May 2024 | Robotics Research Assistant                                                                                                                                                                                                                                          | Technical University Munich, Germany, part-time |
|                     | Working on formally verified safe physical human-robot interaction for the TUM CPS group. Implementing kinematics and trajectory planning algorithms and contributing to a publication accepted in IEEE Transactions on Robotics.                                    |                                                 |
| Nov 2020 - Apr 2021 | CS Teaching Assistant                                                                                                                                                                                                                                                | Technical University Munich, Germany, part-time |
|                     | Teaching introductory computer science with a focus on object-oriented programming in Java and algorithmic problem-solving, and grading weekly assignments and exams.                                                                                                |                                                 |

Education

|                     |                                                                                                                                                                                                                         |                                      |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Apr 2023 - Sep 2026 | M.Sc. Robotics, Cognition, Intelligence                                                                                                                                                                                 | Technical University Munich, Germany |
|                     | Interdisciplinary degree spanning computer science, electrical engineering, and mechanical engineering with specialization in robotics, computer vision, and machine learning.                                          |                                      |
| Feb 2024 - Aug 2024 | Siemens Mentoring Programme                                                                                                                                                                                             | Siemens, Munich, Germany             |
|                     | Career development programme for high-performing students. Presented the project "Smart 3D Printing with Machine Learning" at the Siemens RIE Conference.                                                               |                                      |
| Sep 2019 - Mar 2023 | B.Sc. Informatics                                                                                                                                                                                                       | Technical University Munich, Germany |
|                     | Computer science degree with focus on cyber-physical systems and artificial intelligence. Bachelor's thesis: <i>Provably Safe Human-Robot Interaction for Manipulation using Power and Force Limiting</i> (Grade: 1.3). |                                      |

Projects

|                |                                                                                                                                                                                                      |                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Feb 2026       | Hack-Nation Global AI Hackathon                                                                                                                                                                      | MIT & CDTM                             |
|                | Built "EasyBuy," an AI e-commerce assistant. Invited to the Venture Lab programme (15–20 teams chosen from 6,000+ applicants).                                                                       |                                        |
| Dec 2025       | RoboTUM Hackathon                                                                                                                                                                                    | RoboTUM & Motius, Munich, Germany      |
|                | Built a full robot learning pipeline for an SO-101 platform with LeRobot, including teleoperation data collection, fine-tuning of ACT and SmoVLA, and real-time inference for a pick-and-place task. |                                        |
| Since Nov 2025 | AI Grid                                                                                                                                                                                              | National AI Research Network (Germany) |
|                | Participating in a research-oriented AI network connecting academia and industry to advance applied artificial intelligence.                                                                         |                                        |